Highly accurate protein structure prediction with AlphaFold

Presentor:

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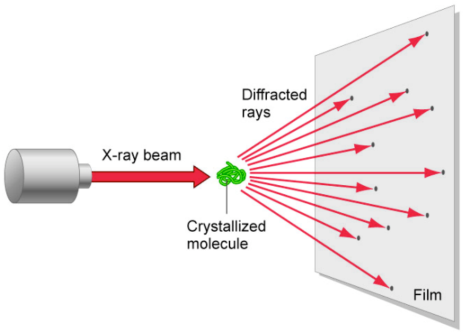
Contents

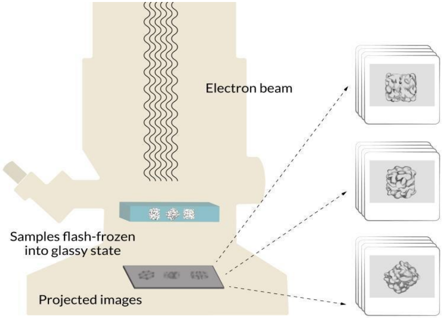
Proteins

- Essential for life, and their function depends on its 3D structure

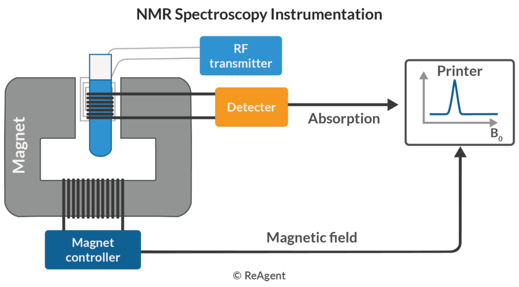
- Experimental determination of protein structure is **Slow** & **Costly**
  - Resulting in structures for only a small fraction of known protein sequences



X-ray crystallography



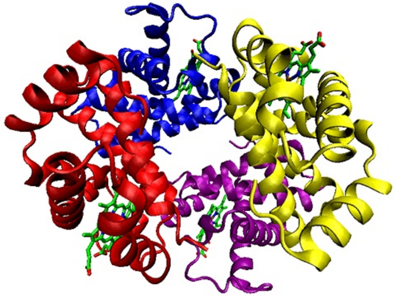
Cryogenic EM



NMR Spectroscopy

Background & Motivation

Predicting 3D structures of Protein



- Open challenge for over 50 years

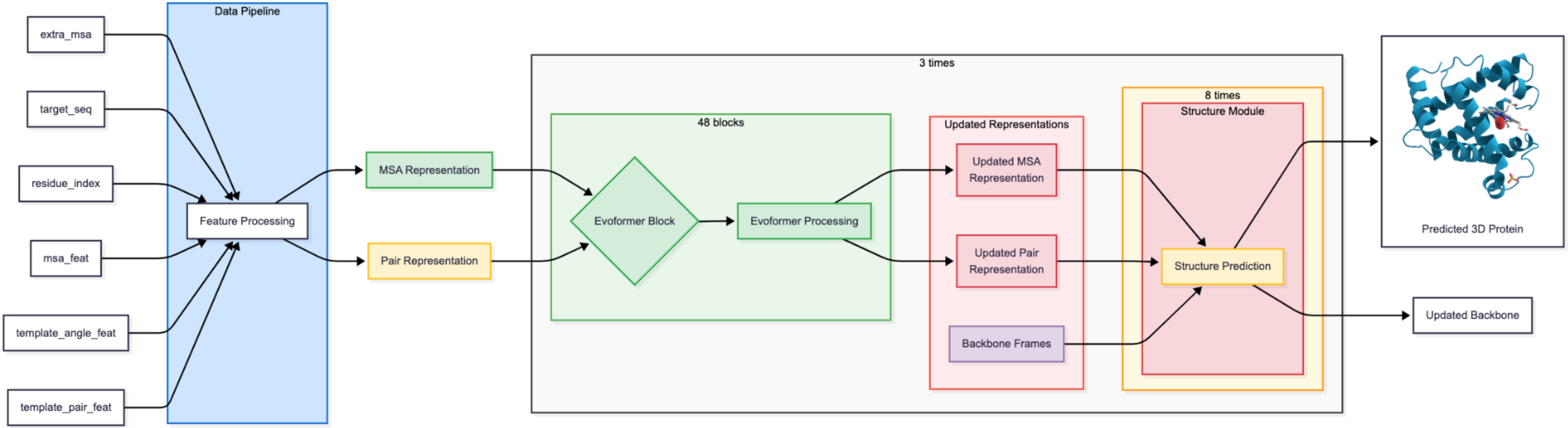
- Existing methods
  - Physical (Physics-Based) Approaches
  - Evolutionary (Bioinformatics-Based) Approaches

Background & Motivation

CASP

- In 1994, John Mault started a competition called CASP (Critical Assessment of Structure Prediction)
- Goal is to design a computer model that take a sequence of amino acid output a structure
- Perfect Prediction 100% accuracy, 90% and above is fine enough model.
- In 2018 CASP13, A7D achieve ~68% accuracy , Top one
- In 2020 CASP14 , AlphaFold2 achieved ~87%

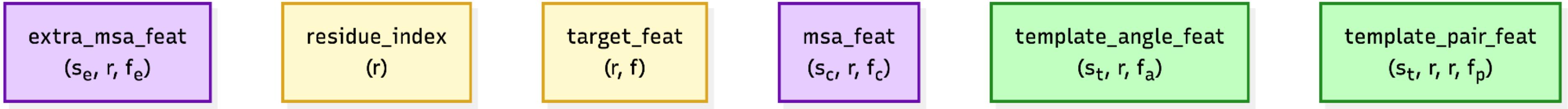
The AlphaFold



Data Pipeline

The Data Pipeline is the "Pre-processing" stage

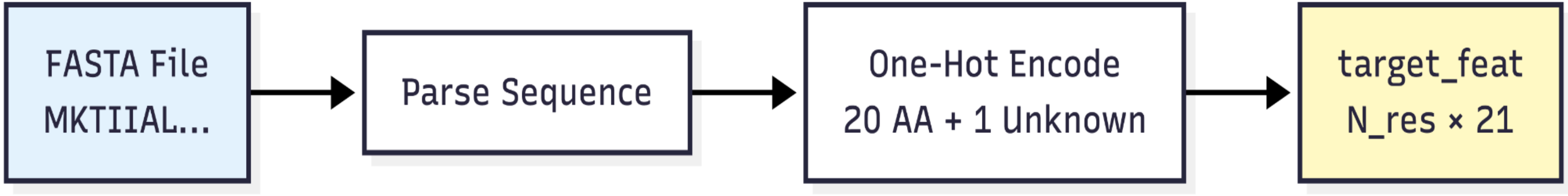
INPUTS OF ALPHAFOLD

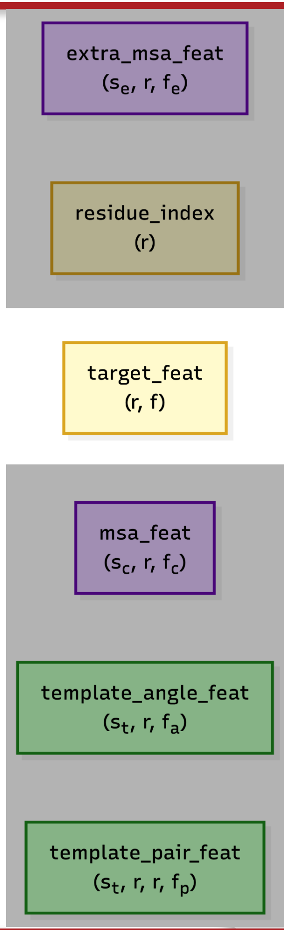


The AlphaFold

Data Pipeline

target_feat





Data Pipeline

residue_index

Protein Length N_res → Generate Array 0, 1, 2, ..., N_res-1 → residue_index N_res

Residue index	:	1	2	3	4	5	6	7	8	9	10
Sequence	:	A	B	C	D	E	F	G	H	I	J
Exp. 3D Shape	:	Y	Y	Y	N	N	Y	Y	Y	Y	Y
pLDDT Score	:	H	H	H	L	L	H	H	H	H	H

extra_msa_feat (s_e, r, f_e)

residue_index (r)

target_feat (r, f)

msa_feat (s_c, r, f_c)

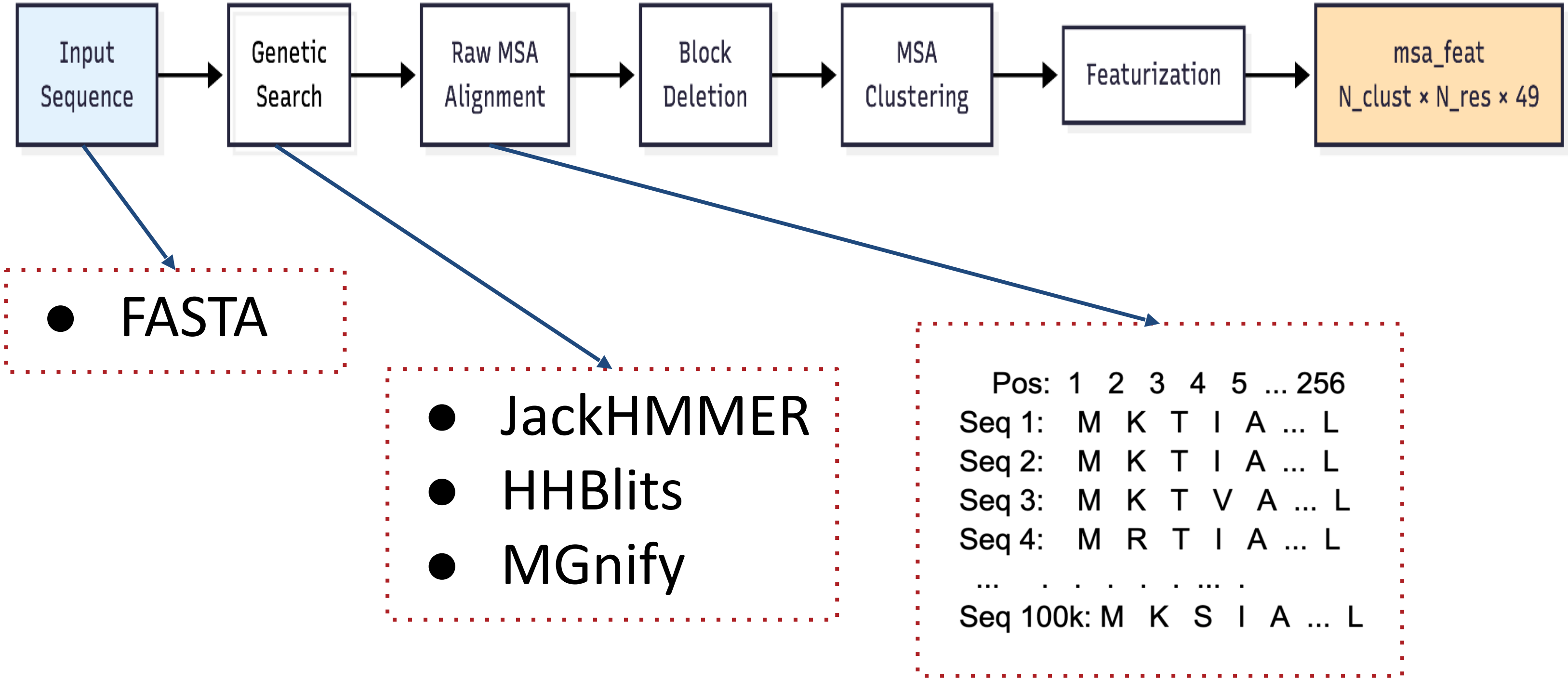
template_angle_feat (s_t, r, f_a)

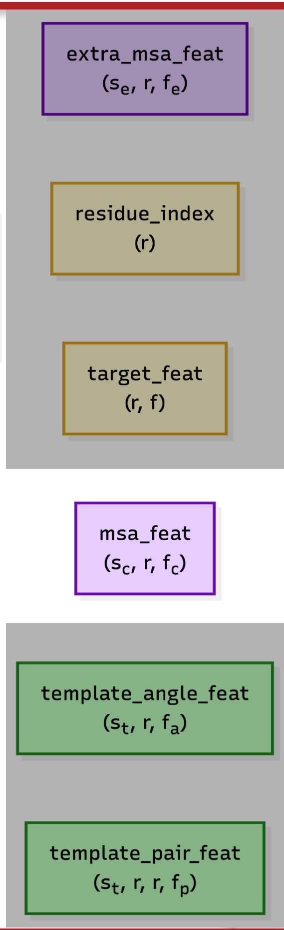
template_pair_feat (s_t, r, r, f_p)

The AlphaFold

Data Pipeline

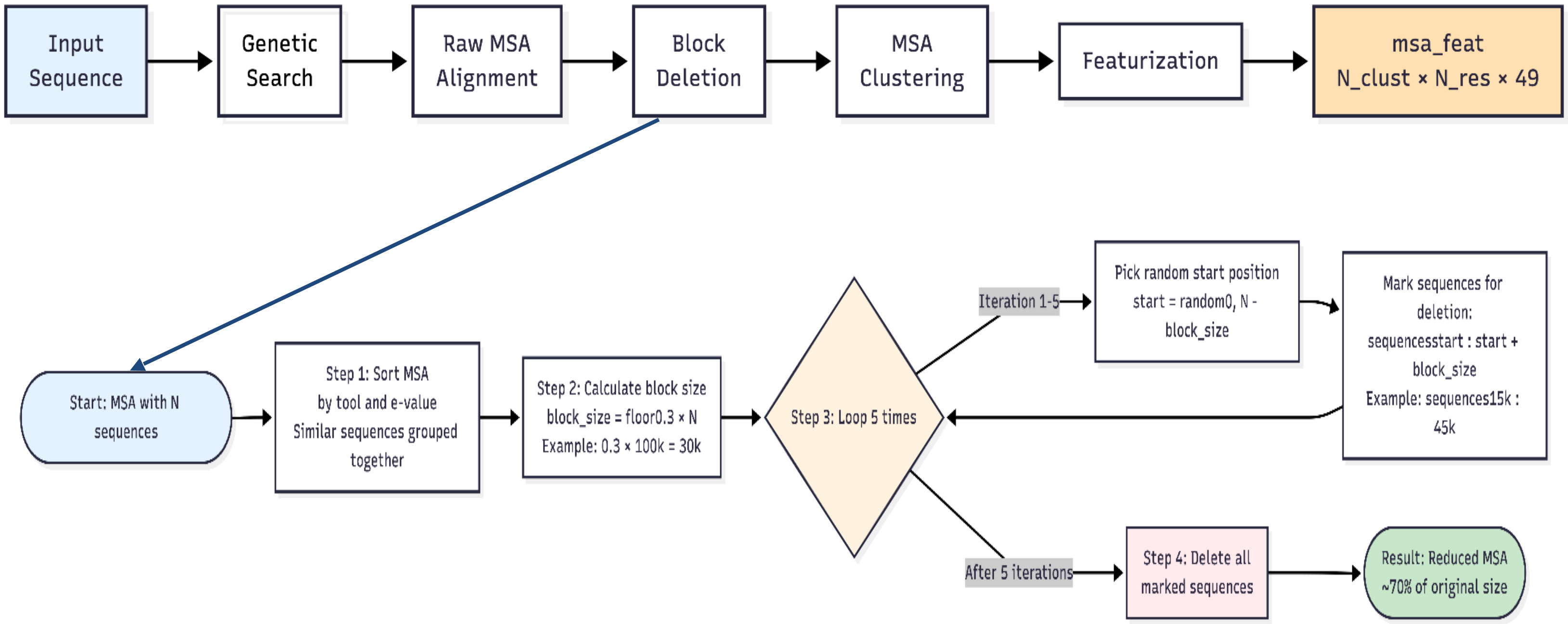
msa_feat

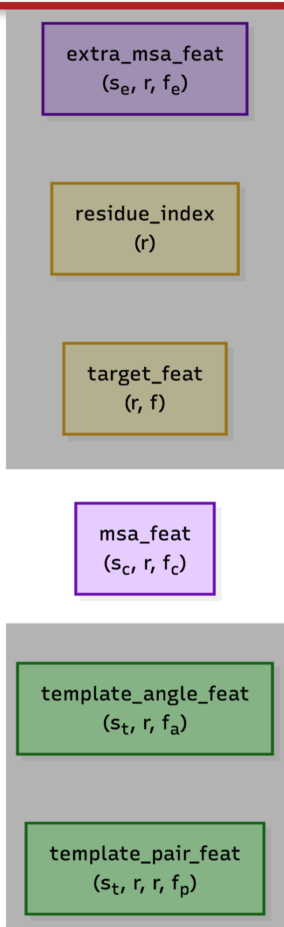




Data Pipeline

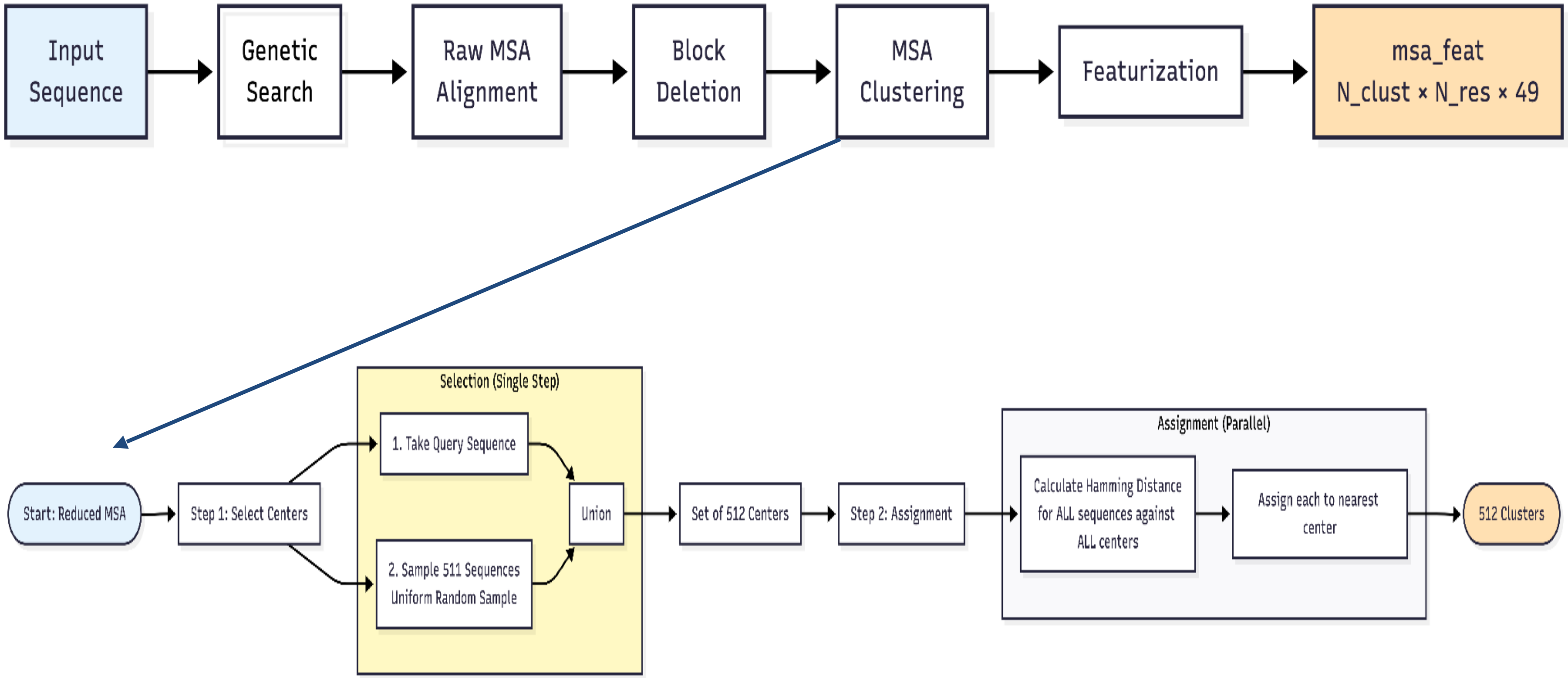
msa_feat

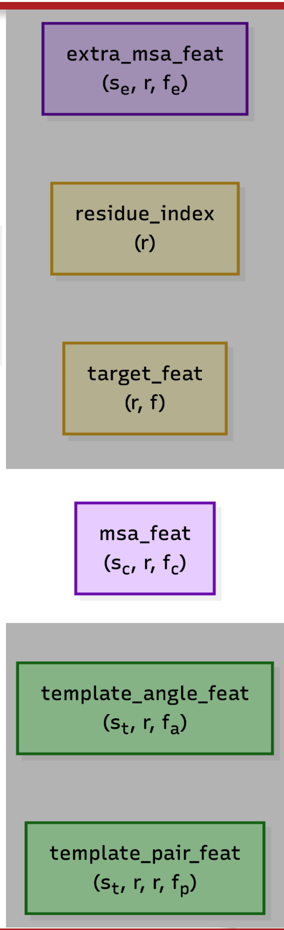




Data Pipeline

msa_feat

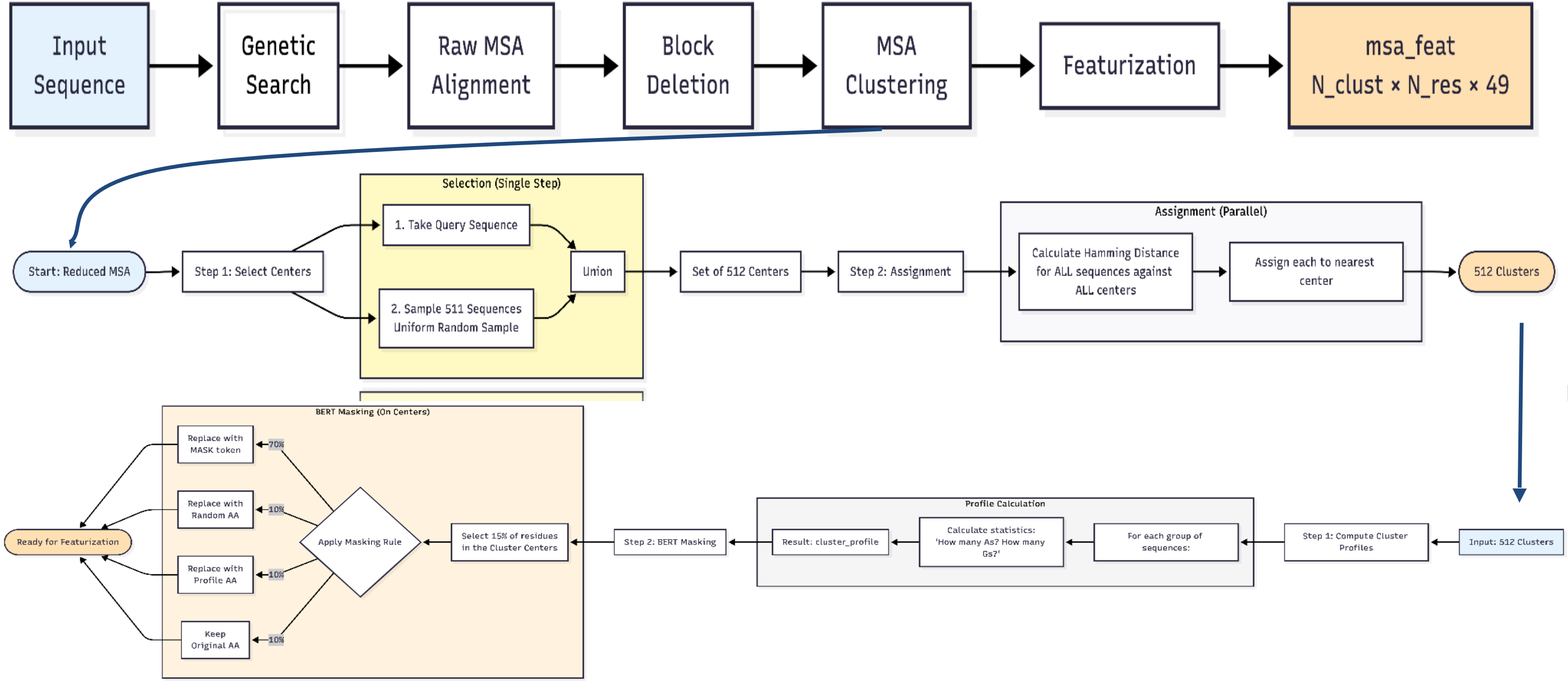




The AlphaFold

Data Pipeline

msa_feat



extra_msa_feat
(s_e, r, f_e)

residue_index
(r)

target_feat
(r, f)

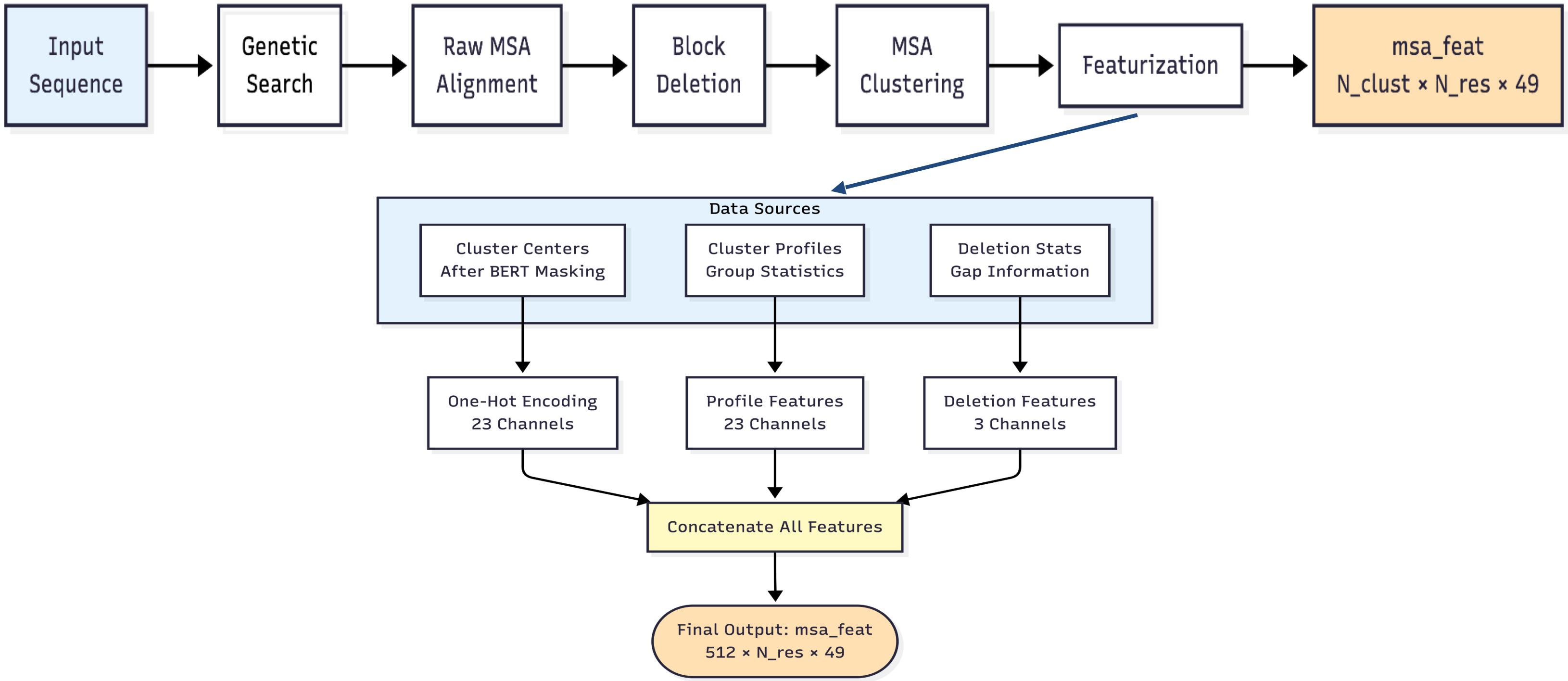
msa_feat
(s_c, r, f_c)

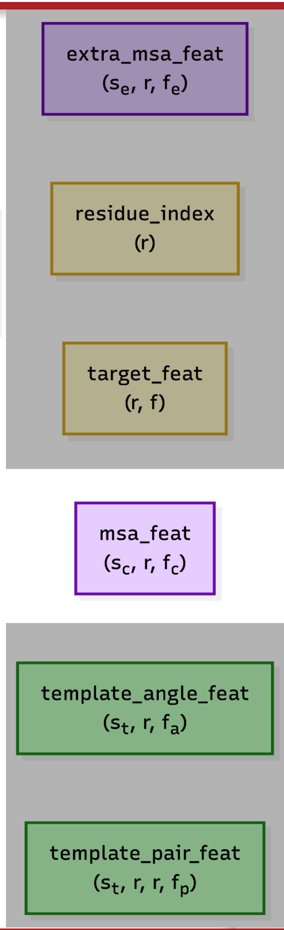
template_angle_feat
(s_t, r, f_a)

template_pair_feat
(s_t, r, r, f_p)

Data Pipeline

msa_feat

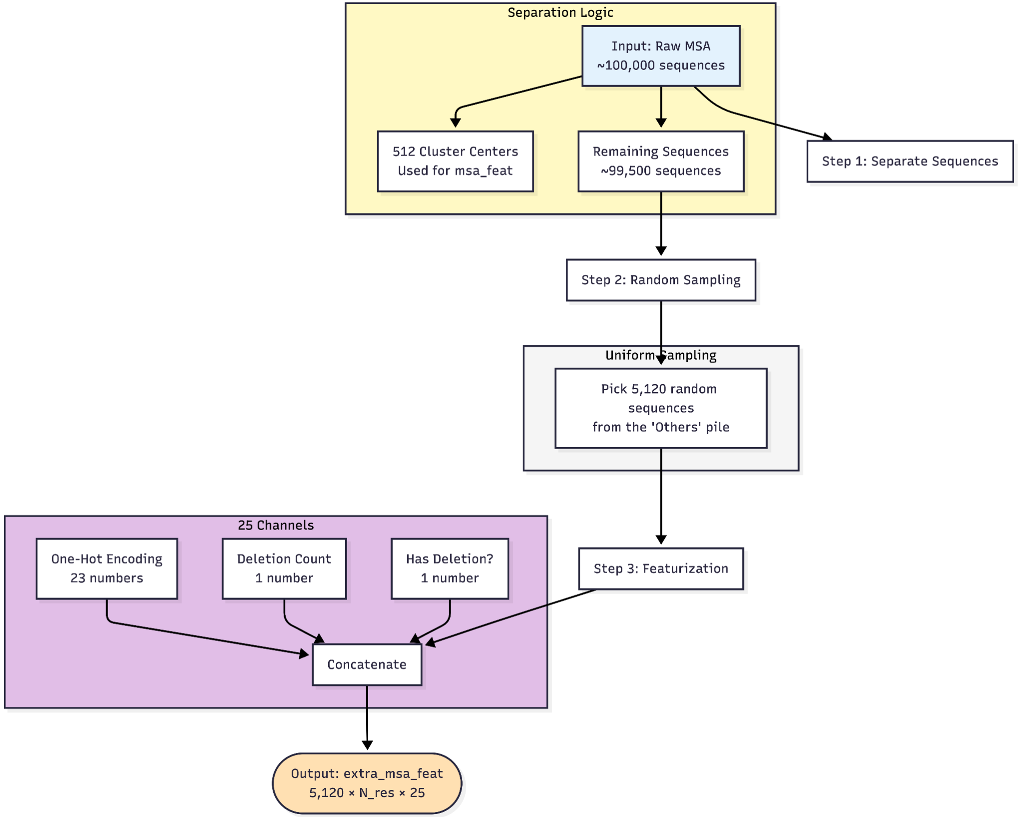




The AlphaFold

Data Pipeline

extra_msa_feat



extra_msa_feat
(s_e, r, f_e)

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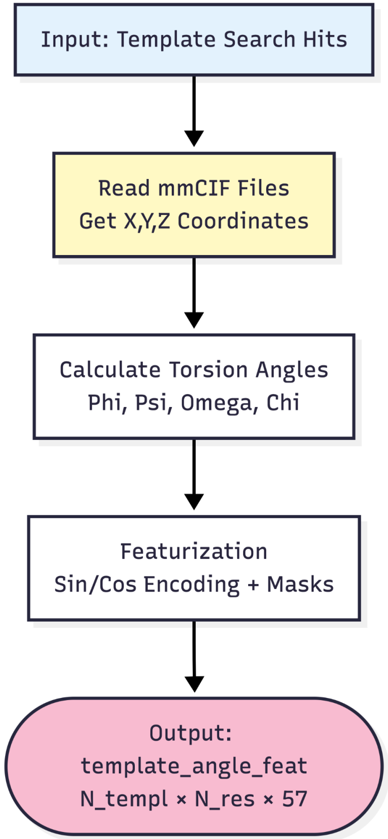
template_angle_feat
(s_t, r, f_a)

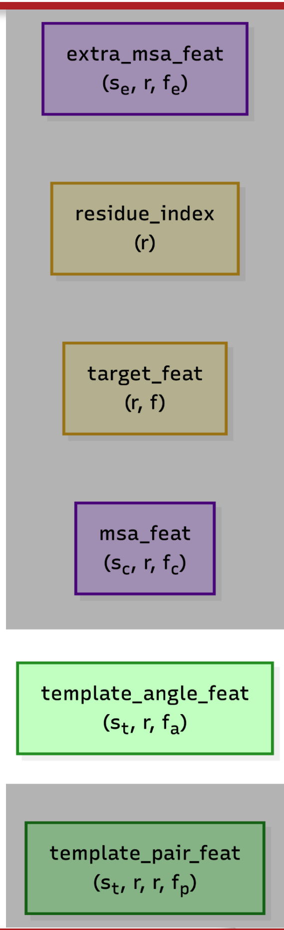
template_pair_feat
(s_t, r, r, f_p)

Data Pipeline

template_angle_feat



- Phi (φ): Twist between N and Cα atoms.
- Psi (ψ): Twist between Cα and C atoms.
- Omega (ω): Twist between C and N atoms
- Chi (χ): Twists in the sidechain (Cα-Cβ, etc.)



The AlphaFold

Data Pipeline

template_pair_feat

Input: Template Hits

↓

Read mmCIF Files
Get X,Y,Z Coordinates

↓

Calculate Geometry

↓

Featurization
Distogram + Unit Vectors

↓

Output: template_pair_feat
N_templ × N_res × N_res × 88

Pairwise Calculations

- Distances: Cα-Cα distance
- Vectors: Direction between atoms
- Frames: Relative Orientation

extra_msa_feat
(s_e, r, f_e)

residue_index
(r)

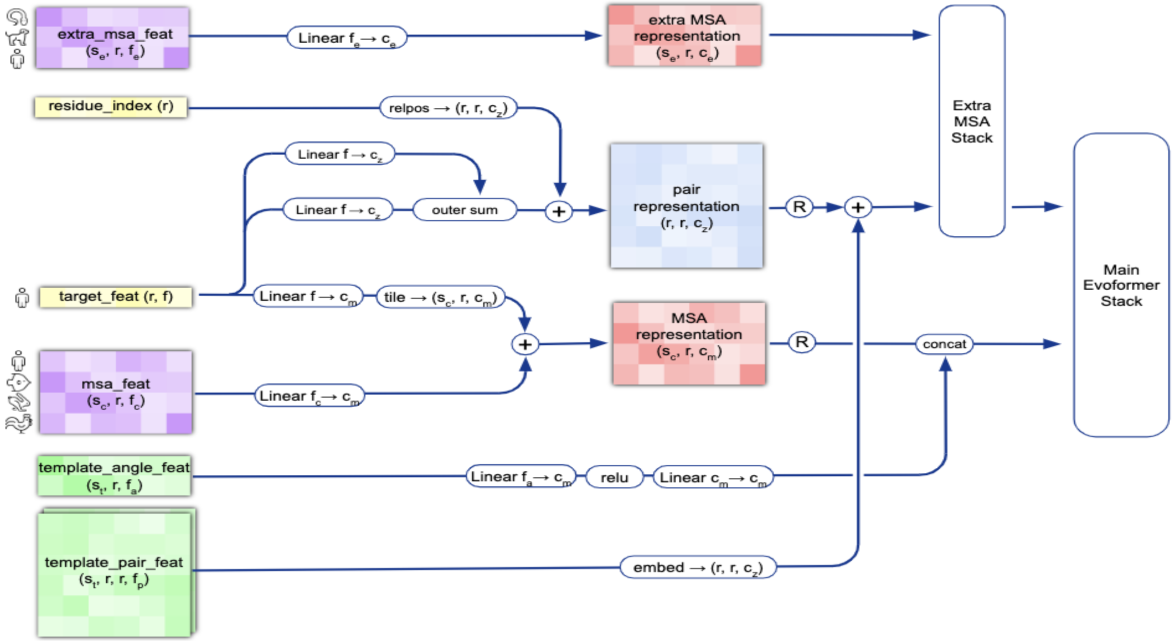
target_feat
(r, f)

msa_feat
(s_c, r, f_c)

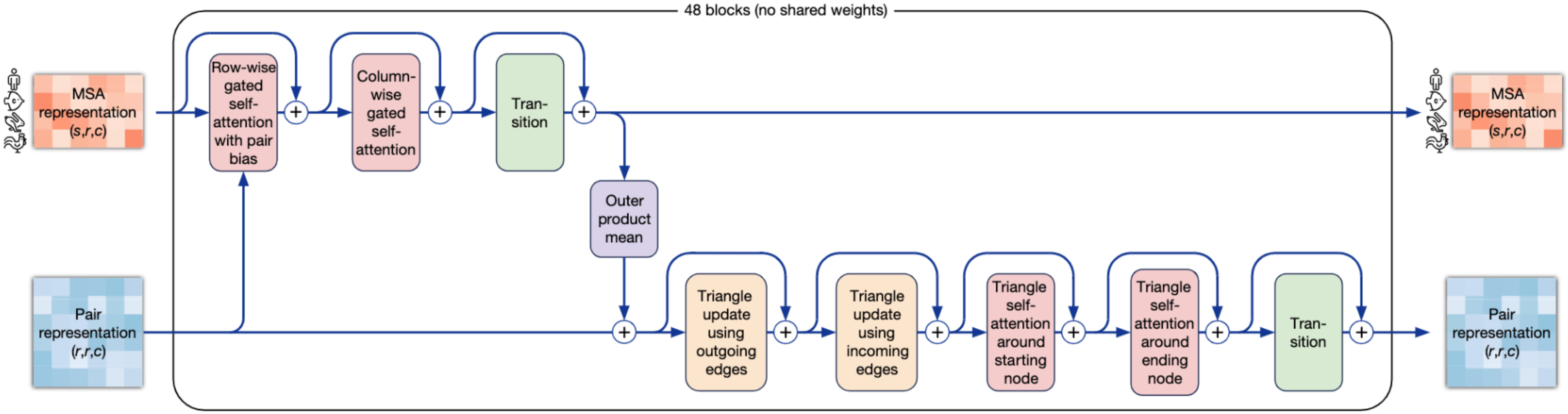
template_angle_feat
(s_t, r, f_a)

template_pair_feat
(s_t, r, r, f_p)

Data Pipeline

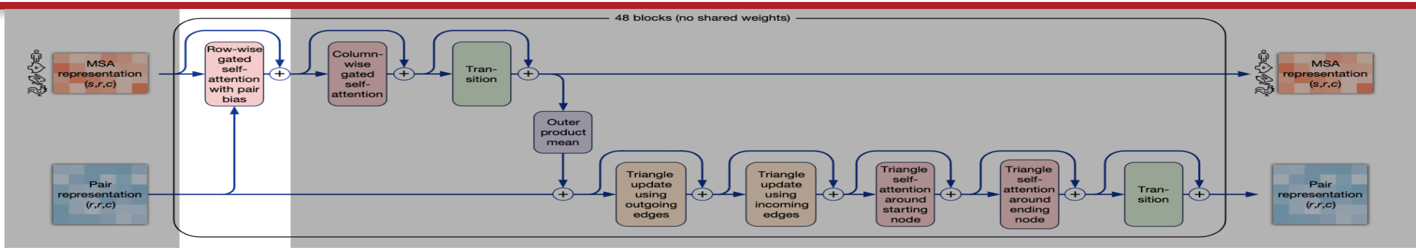


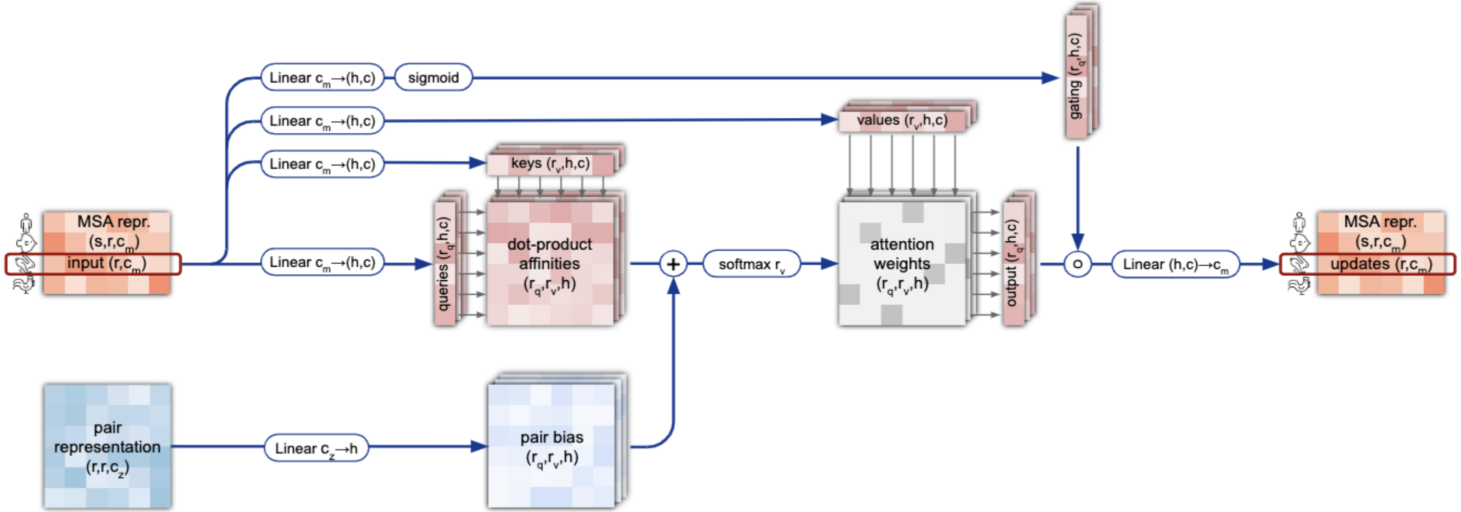
Evoformer


48 blocks (no shared weights)
MSA representation (s,r,c)
Row-wise gated self-attention with pair bias
Column-wise gated self-attention
Tran-sition
Outer product mean
Pair representation (r,r,c)
Triangle update using outgoing edges
Triangle update using incoming edges
Triangle self-attention around starting node
Triangle self-attention around ending node
Tran-sition
MSA representation (s,r,c)
Pair representation (r,r,c)

Evoformer

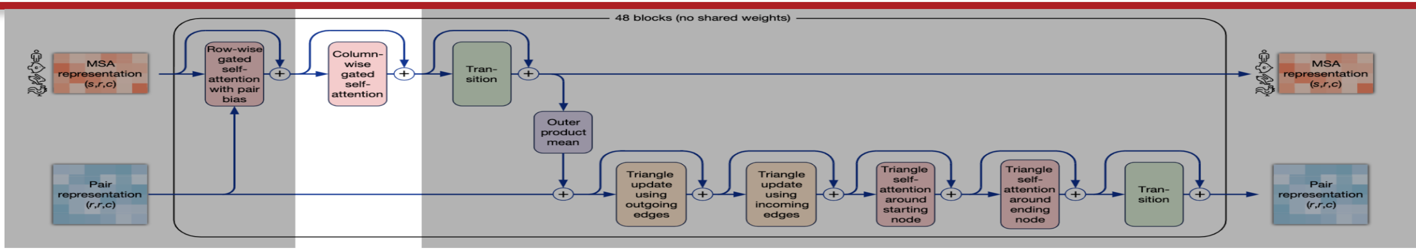
Row-wise gated self- attention

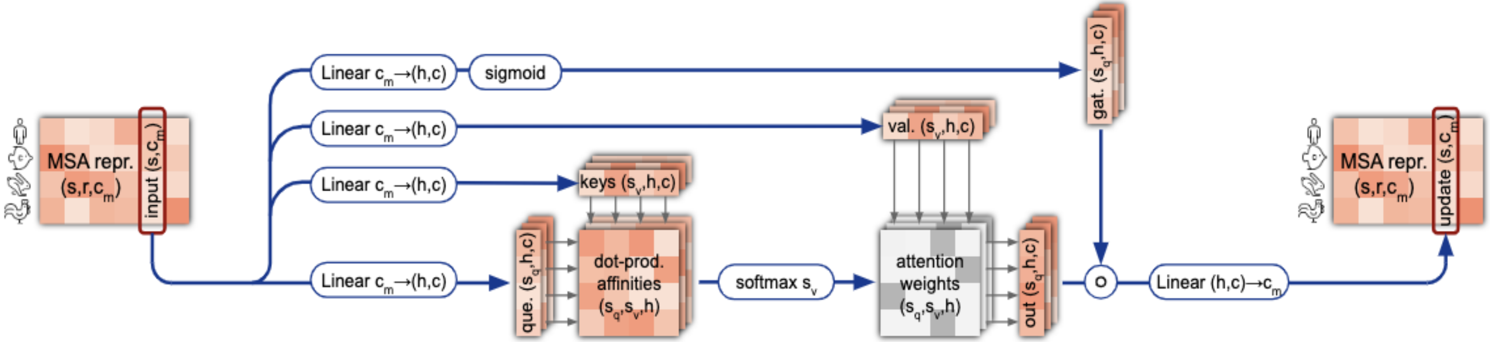




Evoformer

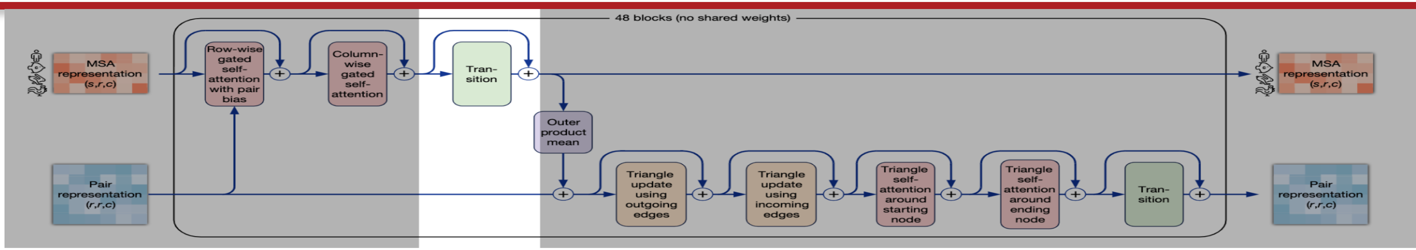
Column-wise gated self-attention

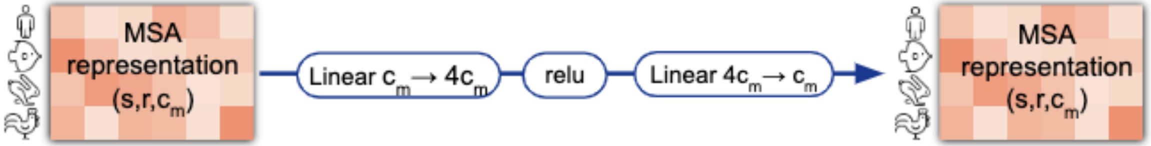




Evoformer

MSA transition

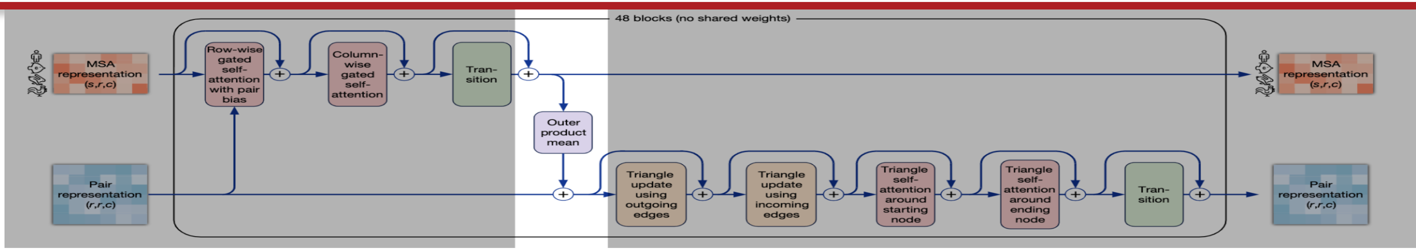


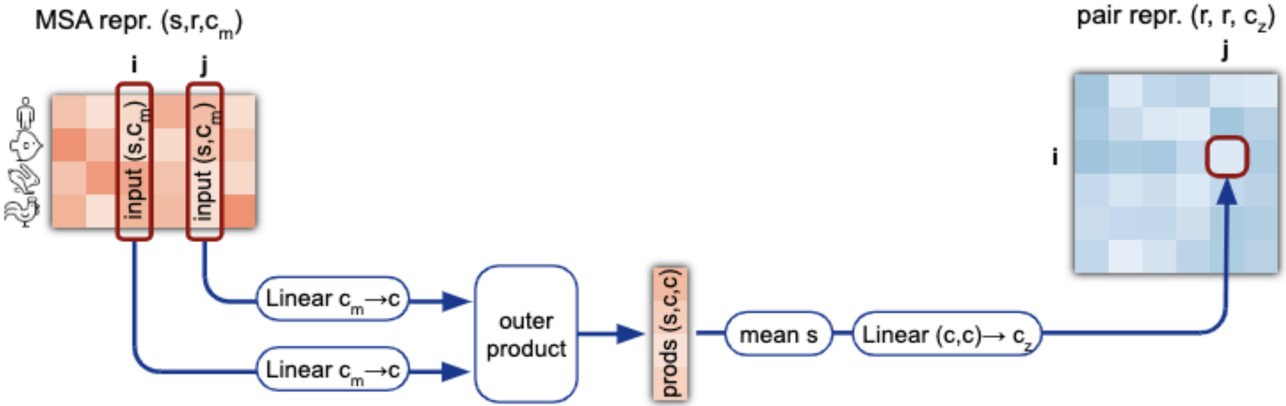


The AlphaFold

Evoformer

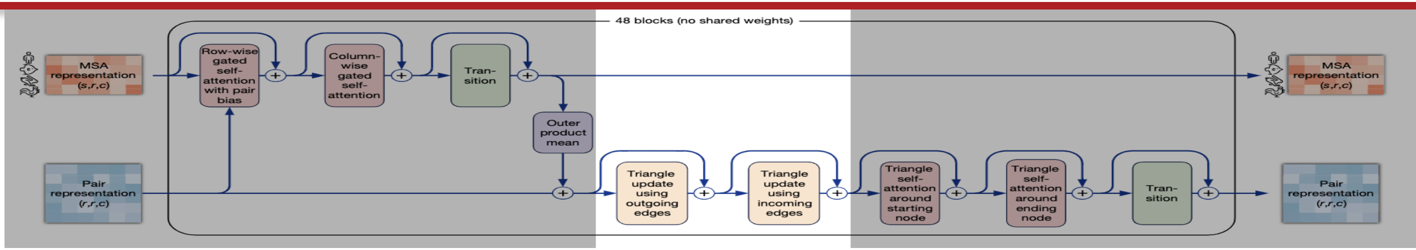
Outer product mean

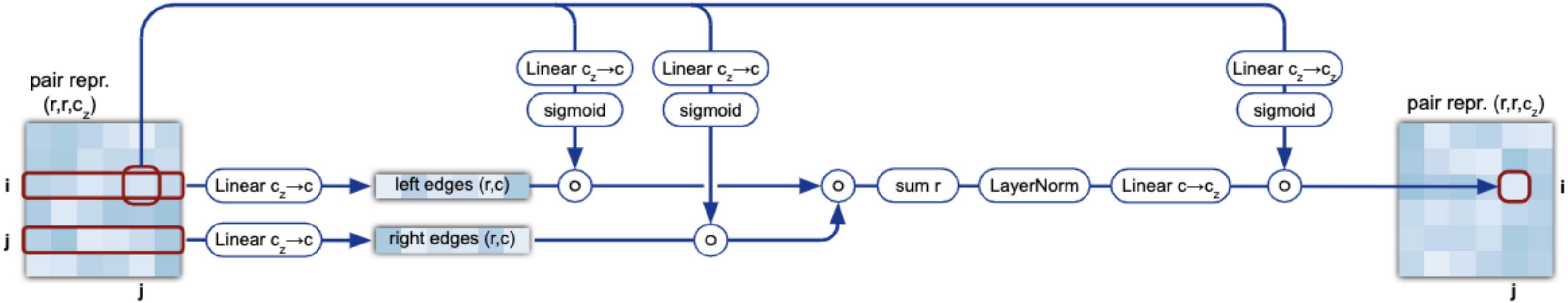




Evoformer

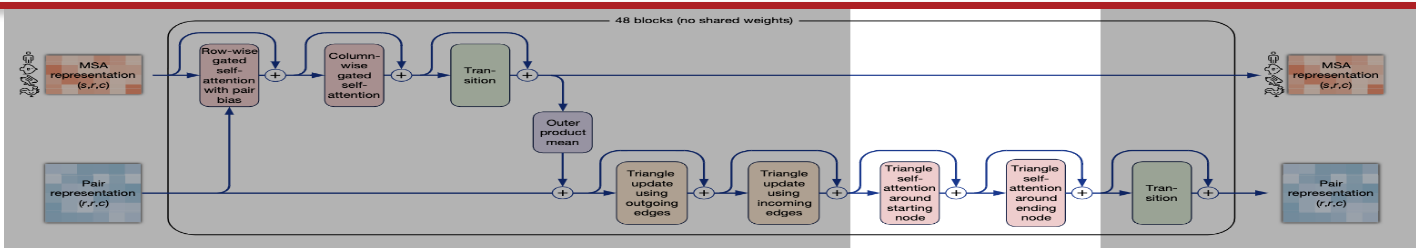
Triangle updates

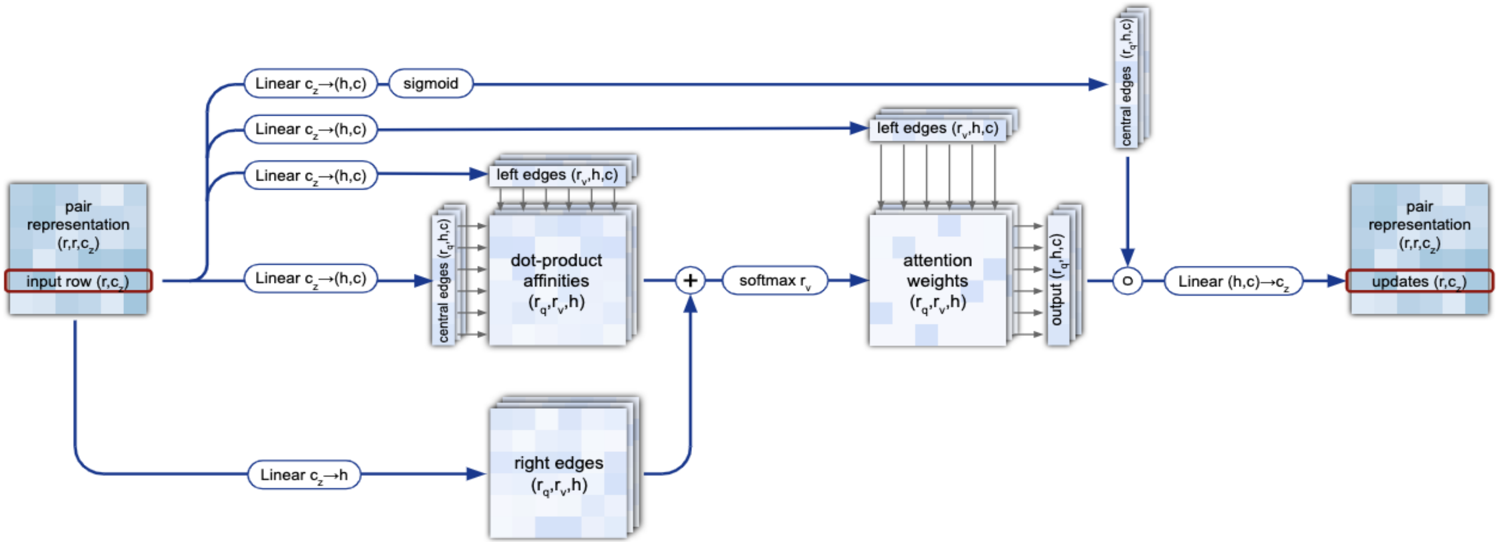




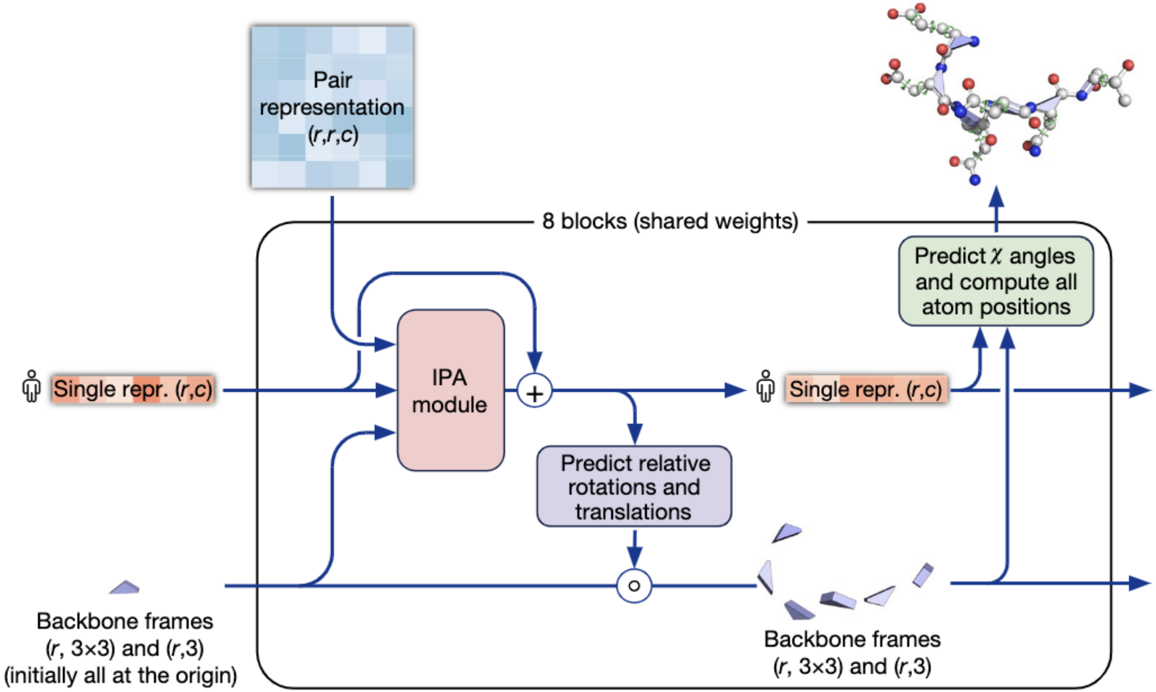
Evoformer

Triangular self-attention



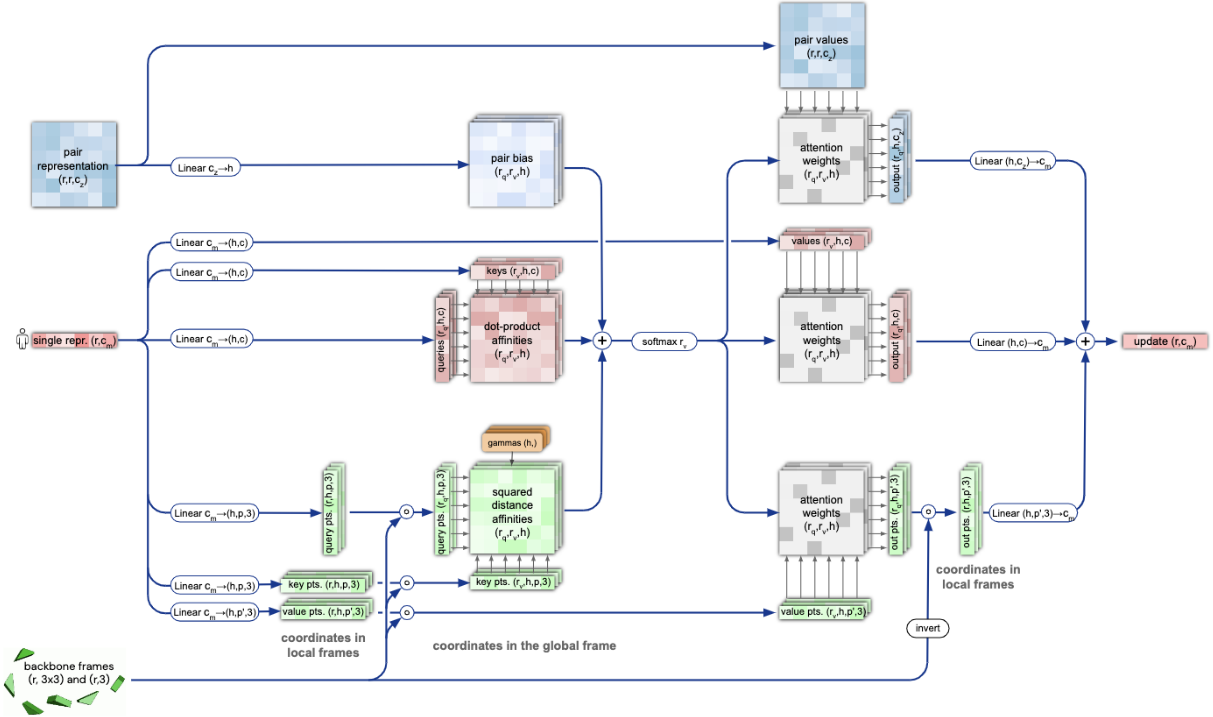


Structure Module

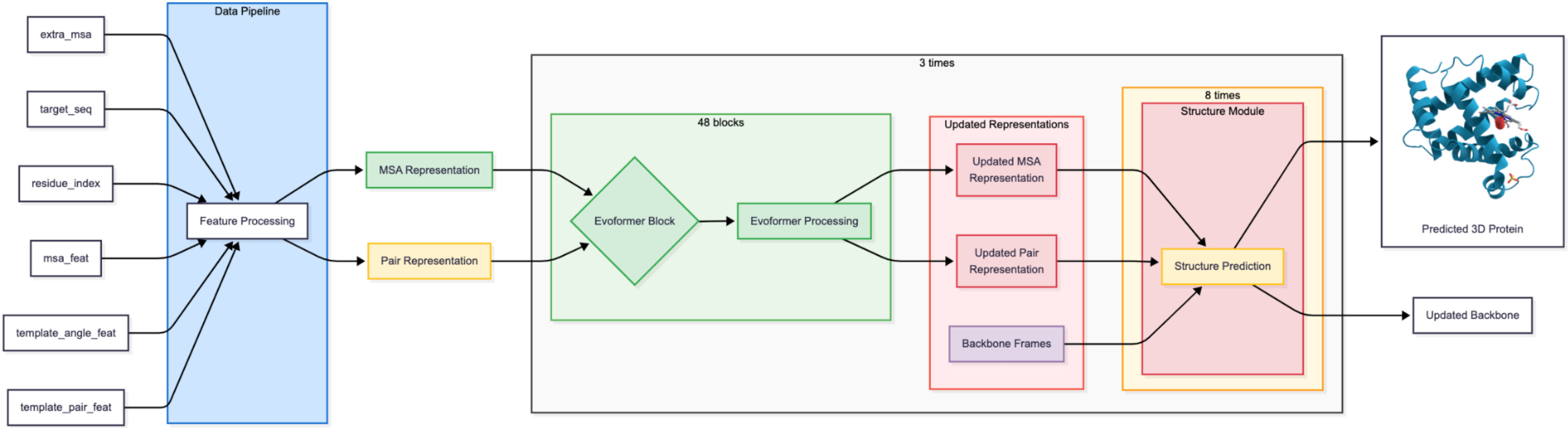


Structure Module

Invariant Point Attention (IPA)



The AlphaFold



Error Calculation

Loss function

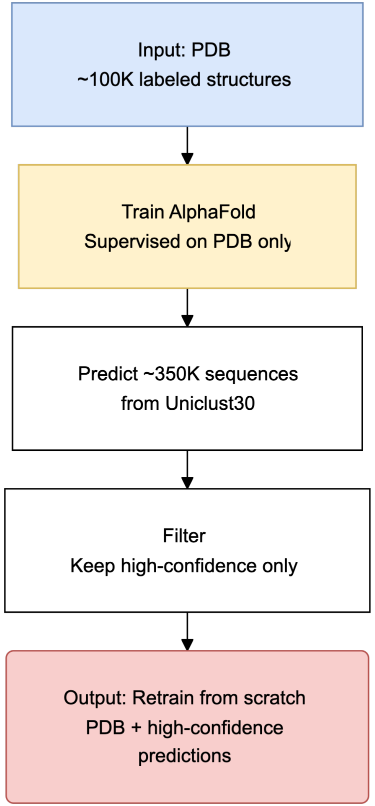
$$\mathcal{L} = \begin{cases} 0.5\mathcal{L}_{\text{FAPE}} + 0.5\mathcal{L}_{\text{aux}} + 0.3\mathcal{L}_{\text{dist}} + 2.0\mathcal{L}_{\text{msa}} + 0.01\mathcal{L}_{\text{conf}} & \text{training} \\ 0.5\mathcal{L}_{\text{FAPE}} + 0.5\mathcal{L}_{\text{aux}} + 0.3\mathcal{L}_{\text{dist}} + 2.0\mathcal{L}_{\text{msa}} + 0.01\mathcal{L}_{\text{conf}} + 0.01\mathcal{L}_{\text{exp resolved}} + 1.0\mathcal{L}_{\text{viol}} & \text{fine-tuning} \end{cases},$$

- L_{FAPE} = Frame Aligned Point Error Loss
- L_{aux} = is a loss from auxiliary metrics in structure module
- L_{dist} = is loss from distogram prediction
- L_{msa} = is a loss from BERT-like masked MSA position prediction
- L_{conf} = is a loss from model confidence in pLDDT prediction.
- $L_{exp\ resolved}$ = is a loss from a head, predicting, if structure comes from a highly accurate experimental prediction, such as CryoEM or high-resolution X-ray crystallography.
- L_{viol} = is a loss for structural violations; penalizes unlikely lengths of bonds, torsion angles and sterically clashing atoms, even if ground truth structure is unavailable.

Engineering Tricks

Self-distillation

Noisy Student Training

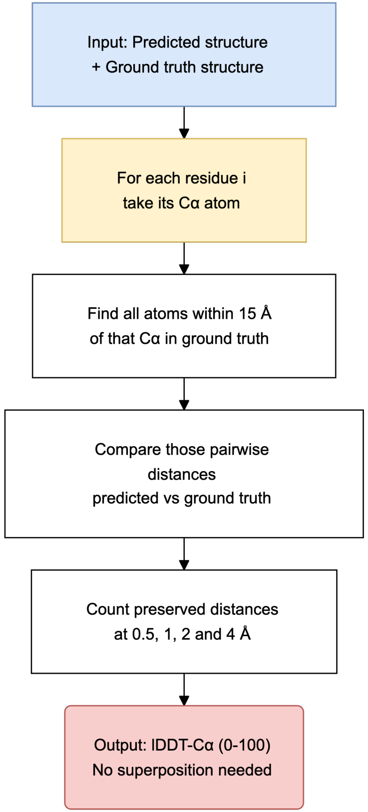


- Labeled data: under 200K structures in the PDB
- Unlabeled data: over 200M known sequences
- Semi-supervised learning, using **Noisy Student Training**
- Cropping and MSA subsampling stop the network simply copying its own predictions
- In the Fig. 4a ablation it is the only change scoring above the baseline

Engineering Tricks

pLDDT

predicted Local Distance Difference Test



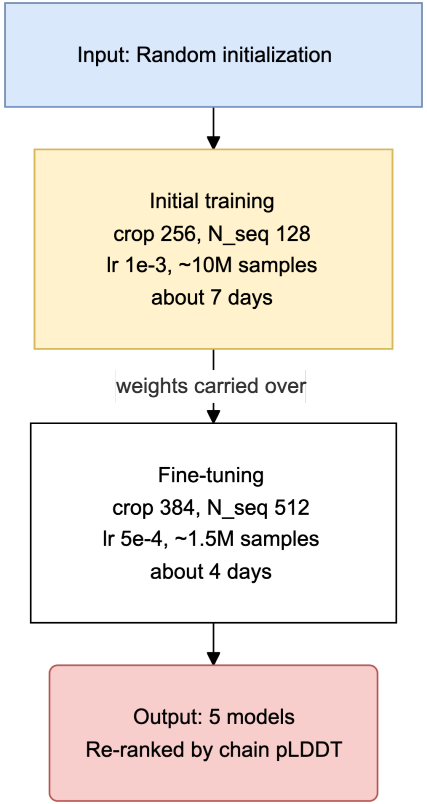
Predicting the score without the answer

- IDDT needs the true structure, so it cannot be computed at prediction time
- pLDDT predicts it from the final single representation s_i
- Projected to 50 bins, each covering 2% of IDDT-Cα
- Training: cross-entropy against the true bin → L_conf
- Inference: expected value of the bins → 0-100
- IDDT-Cα = 0.997 × pLDDT − 1.17 (r = 0.76, n = 10,795 chains)
- >90 very high · 70-90 confident · 50-70 low · <50 likely disordered

Engineering Tricks

Training protocol

Two stages on 128 TPUv3 cores



- Trained on TPUv3 with a mini-batch of 128, one example per core
- 16 GiB per core is too little for 48 blocks, so activations are recomputed rather than stored
- Violation and “experimentally resolved” losses are switched on for fine-tuning only
- About two weeks end to end, including building the distillation set

Thank You!